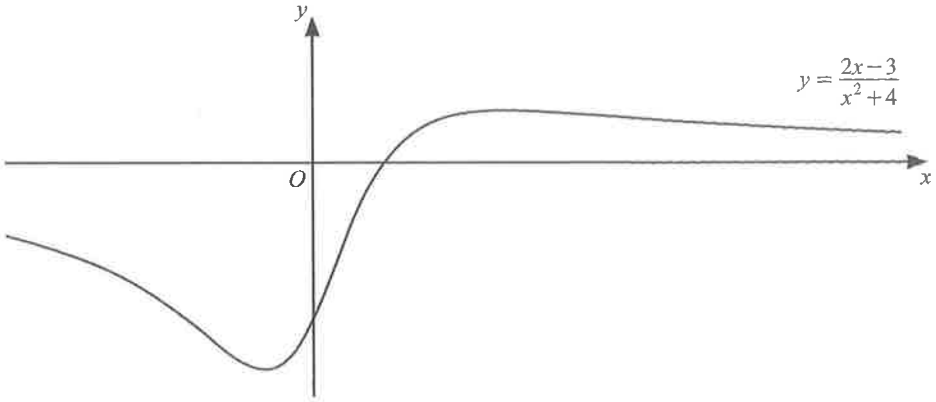
Unit 10 Gradients, Derivatives and Differentiation Techniques

Basic

1.



The diagram shows part of the graph of $y = \frac{2x-3}{x^2+4}$. Find the values of x for which y is increasing. [4]

[N20/I/4]

2. The function f is defined, for all values of x, by

$$f(x) = x^2(1 - x).$$

Find the values of x for which f is an increasing function. [3]

[N15/I/1]

Intermediate

3. The equation of a curve is $y = x^3 - ax^2 + bx + 4$, where a and b are constants.

(a) Show that y is always increasing if $a^2 < 3b$. [3]

(b) In the case where $a = 8$ and $b = 10$, find the x-coordinate of each of the three points at which the curve intersects the x-axis. [4]

[N22/I/9]